Affine Dual Jacobi–Trudi Identities for Classical Hall–Littlewood Polynomials via Complementary Minors

Abstract

We prove the six affine dual Jacobi–Trudi identities conjectured by Warnaar for rectangular Hall–Littlewood polynomials of types B_n, C_n, and BC_n, together with his additional type-B_n identity at $s = t$. We give a uniform proof by adapting the complementary-minor method used by Aitken for symmetric-function duality.

Keywords. Hall–Littlewood polynomials; Jacobi–Trudi identities; Weyl characters; complementary minors; affine root systems.

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1 Introduction and main identities

Classical dual Jacobi–Trudi identities express rectangular Weyl characters of rank n as determinants of order k, where k is the width of the rectangle. Warnaar proposed affine analogues for classical Hall–Littlewood polynomials in [7, Conjectures 1.2–1.4], and a further type-B_n identity in [7, Section 6]. We prove these identities, with one necessary correction at width zero. Warnaar established the $k = 1$ cases of the six numbered conjectures and the $t = 1$ specializations (1.6a) and (1.8) [7, Lemmas 4.1–4.4]; his type-A analogue is [7, Theorem 1.1]. The applications to affine and q, t-Rogers–Ramanujan identities developed there use the classical-type formulas proved here.

For $x = (x_1, \dots, x_n)$, define

$$\begin{aligned} \dot{E}(u) &= \prod_{a=1}^{n} (1 + ux_a)(1 + ux_a^{-1}) = \sum_{r=0}^{2n} \dot{e}_r u^r, \\ \ddot{E}(u) &= (1 + u)\dot{E}(u) = \sum_{r=0}^{2n+1} \ddot{e}_r u^r. \end{aligned} \tag{1.1}$$

Both coefficient families are extended by zero to all integer indices. Thus

$$\dot{e}_{2n-r} = \dot{e}_r, \qquad \ddot{e}_{2n+1-r} = \ddot{e}_r, \qquad \ddot{e}_r = \dot{e}_r + \dot{e}_{r-1}.$$

For $1 \le i \le k$ and $m \in \mathbb{Z}$, use the row kernels:

$$\begin{aligned} \Lambda_i(m) &= \dot{e}_{n-i+m} - \dot{e}_{n+i+m}, & h_i(m) &= \dot{e}_{n-i+m} + \dot{e}_{n+i+m-1}, \\ h_i^+(m) &= \dot{e}_{n-i+m} + \dot{e}_{n+i+m-2}, & \widehat{h}_i(m) &= \ddot{e}_{n-i+m} + \ddot{e}_{n+i+m-1}. \end{aligned}$$

For weights $w_j(y)$, kernels $F_i(m)$, and $\mathcal{L} \subseteq \mathbb{Z}^k$, put

$$\mathfrak{J}_K[w, F; \mathcal{L}] := \sum_{(y_1,\dots,y_k)\in\mathcal{L}} \det_{1\le i,j\le k}[w_j(y_i)F_i(j - Ky_i)]. \tag{1.2}$$

In this notation an expression in j, y in the weight slot specifies $w_j(y)$. All sums below are finite by zero extension. Write $\mathbb{Z}^k_{\mathrm{even}} = \{y \in \mathbb{Z}^k : y_1 + \cdots + y_k \in 2\mathbb{Z}\}$. The empty determinant and the sum indexed by the empty vector both have value 1. When $k = 0$, no entry or weight in (1.2) is evaluated. The coefficient field is $\mathbb{Q}(t, x_1, \dots, x_n)$, except in the two square-root cases, where it is $\mathbb{Q}(\tau, x_1, \dots, x_n)$ and $t^{r/2} := \tau^r$ for $r \in \mathbb{Z}$.

Theorem A. *For $n \ge 1$, the following identities hold for $k \ge 0$, except that (1.5) requires $k \ge 1$:*

$$P^{B_n}_{(k^n)}(x; t, 0) = \mathfrak{J}_{2k+1}[(-1)^y t^{(k+\frac12)y^2-(j-\frac12)y}, h; \mathbb{Z}^k], \tag{1.3}$$

$$P^{B_n}_{(k^n)}(x; \tau^2, -\tau) = \mathfrak{J}_{2k}[\tau^{2ky^2-(2j-1)y}, h; \mathbb{Z}^k], \tag{1.4}$$

$$P^{B_n}_{(k^n)}(x; t, 1) = \tfrac12\mathfrak{J}_{2k}[(-1)^y t^{ky^2-(j-1)y}, h^+; \mathbb{Z}^k], \tag{1.5}$$

$$P^{C_n}_{(k^n)}(x; t, 0) = \mathfrak{J}_{2k+2}[t^{(k+1)y^2-jy}, \Lambda; \mathbb{Z}^k], \tag{1.6}$$

$$P^{C_n}_{(k^n)}(x; t, t) = \mathfrak{J}_{2k}[t^{ky^2-jy}, \Lambda; \mathbb{Z}^k_{\mathrm{even}}], \tag{1.7}$$

$$P^{BC_n}_{(k^n)}(x; \tau^2, -\tau, 0) = \mathfrak{J}_{2k+1}[(-1)^y \tau^{(2k+1)y^2-2jy}, \Lambda; \mathbb{Z}^k]. \tag{1.8}$$

At width zero the normalized value in (1.5) is $P^{B_n}_{\varnothing}(x; t, 1) = 1$.

These are, in order, Conjectures 1.2(a)–(c), 1.3(a)–(b), and 1.4 of [7]. The literal $k = 0$ assertion of Conjecture 1.2(c) would give $1/2$ on the right of (1.5); its correct normalized endpoint is therefore separate from the positive-width identity.

Theorem B. *For $n, k \ge 1$,*

$$P^{B_n}_{(k^n)}(x; t, t) = \tfrac12\mathfrak{J}_{2k}[(-1)^y t^{ky^2-(j-1)y}, \widehat{h}; \mathbb{Z}^k]. \tag{1.9}$$

The normalized width-zero value is $P^{B_n}_{\varnothing}(x; t, t) = 1$.

Theorem B is the additional, unnumbered identity in [7, Section 6]. Our Hall–Littlewood conventions are those of [7, (2.16)–(2.17)]; for the type-BC Weyl sum and polynomiality, see also [6, Definition 2.4 and Theorems 2.5–2.6].

The proof uses complementary minors, a familiar source of dual symmetric-function determinants [1, 4]. The corresponding proportionality of complementary maximal minors is also the basic Gale-dual relation; see [5, Lemma 2.10]. We apply it to the kernel pairs $\Phi D = 0$ and $V^{\mathsf{T}}C = 0$. Here $\det(\Phi V)/\Delta$ is a finite Weyl representation, while $D^{\mathsf{T}}C$ is the affine matrix. The coefficient kernels D come from (1.1); the affine kernels C come from reflections of weighted residue classes. Explicit complementary blocks fix all signs and constants.

Section 2 derives the finite Weyl determinants. Section 3 gives the normalized transfer and the three coefficient kernels. Section 4 treats four ordinary cases in two pairs. Section 5 treats the fixed residue, fixed center, and adjacent residues responsible for the three half-determinant formulas. Width zero and the classical specializations are collected in Section 6.

2 Finite Weyl determinants

We abbreviate the seven polynomials by $P^{\star}_{n,k}$, where $\star = C_0, C_t, BC, B_0, B_-, B_+, B_t$ denotes the type and boundary parameter in Theorems A and B. For $(a;t)_m = \prod_{r=0}^{m-1}(1-at^r)$, set

$$V_n(t) = \prod_{r=1}^{n}(1+t+\cdots+t^{r-1}).$$

The rectangular Weyl-sum definition, including width zero, is

$$P^{\star}_{n,k} = \frac{1}{V_n(t)\nu_{\star}(k)} \sum_{w\in W_n} w\left[\prod_{a=1}^{n} x_a^{-k}\rho_{\star}(x_a)\prod_{a<b}\frac{(tx_a-x_b)(1-tx_ax_b)}{(x_a-x_b)(1-x_ax_b)}\right], \tag{2.1}$$

where $W_n = S_n \ltimes \{\pm 1\}^n$ acts by permutations and inversions, $\nu_{\star}(k) = 1$ for $k \geq 1$, and the remaining data are as follows. In the BC and B_- rows, $t = \tau^2$.

Case $\star$	$\rho_{\star}(z)$	$\nu_{\star}(0)$
C_0	$(1-z^2)^{-1}$	1
C_t	$(1-tz^2)/(1-z^2)$	$(-t;t)_n$
BC	$(1+\tau z)/(1-z^2)$	1
B_0	$(1-z)^{-1}$	1
B_-	$(1+\tau z)/(1-z)$	$(\tau;\tau^2)_n$
B_+	1	$(-1;t)_n$
B_t	$(1-tz)/(1-z)$	$(-t;t)_n$

Indeed, the symmetric-group normalization of either (k^n), $k>0$, or $\varnothing$ is $V_n(t)$; only the boundary Pochhammer factor changes at width zero.

The standard symmetrizer identity

$$\sum_{\sigma\in S_n}\prod_{a<b}\frac{tz_{\sigma(a)}-z_{\sigma(b)}}{z_{\sigma(a)}-z_{\sigma(b)}} = V_n(t)$$

follows by alternation and comparison of degrees; setting $z_n = 0$ gives the recurrence $V_n = (1+t+\cdots+t^{n-1})V_{n-1}$. Since the other factors in (2.1) are symmetric, it reduces that expression to

$$P^{\star}_{n,k} = \frac{1}{\nu_{\star}(k)} \sum_{\eta\in\{\pm1\}^n}\prod_{a} u_a^{-k}\rho_{\star}(u_a)\prod_{a<b}\frac{1-tu_au_b}{1-u_au_b}, \qquad u_a = x_a^{\eta_a}. \tag{2.2}$$

2.1 Alternants and finite foldings

Write $N_t(z) = \prod_{a<b}(z_a-z_b)(1-tz_az_b)$ and put

$$f_1(z) = z^{n-1}, \qquad f_r(z) = z^{n-r} + t^{r-1}z^{n+r-2} \quad (2\leq r\leq n).$$

The finite numerator calculations reduce to three alternants.

Lemma 2.1. *With rows indexed by $1 \le a \le n$ and columns by $1 \le r \le n$,*

$$\det[f_r(z_a)] = N_t(z), \tag{2.3}$$

$$\det[z_a^{n-r} - t^r z_a^{n+r}] = \prod_a (1 - tz_a^2) N_t(z), \tag{2.4}$$

$$\det[z_a^{n-r} + \tau^{2r-1} z_a^{n+r-1}] = \prod_a (1 + \tau z_a) N_{\tau^2}(z). \tag{2.5}$$

Proof. Each determinant is alternating. On setting $z_b = (tz_a)^{-1}$, rows a, b are proportional, with multipliers $t^{1-n} z_a^{2-2n}$, $-t^{-n} z_a^{-2n}$, and $\tau^{1-2n} z_a^{1-2n}$, respectively; in the last expression $t = \tau^2$. This gives the factors $1 - tz_a z_b$. The second determinant has row factor $1 - tz_a^2$, and the third vanishes at $z_a = -\tau^{-1}$. Divisibility can be checked over the fraction field in the remaining variables and returned to the polynomial ring by Gauss's lemma. The factors are pairwise coprime and, together with $z_a - z_b$, have total degrees

$$3\binom{n}{2}, \qquad 2n + 3\binom{n}{2}, \qquad n + 3\binom{n}{2},$$

which are the respective upper bounds on the determinant degrees. In each case the lowest-degree homogeneous part is $\det[z_a^{n-r}] = \prod_{a<b}(z_a - z_b)$, fixing the constant at 1. □

Set $\phi_m(z) = z^m - z^{-m}$, $\psi_m(z) = z^m - z^{1-m}$, and $C_m(z) = z^m + z^{-m}$. For $L = n + k$ and standard basis $\mathbf{e}_1, \dots, \mathbf{e}_L$, use the foldings

$$[m]_C = \begin{cases} \mathbf{e}_m & m > 0, \\ 0 & m = 0, \\ -\mathbf{e}_{-m} & m < 0, \end{cases} \quad [m]_B = \begin{cases} \mathbf{e}_m & m \ge 1, \\ -\mathbf{e}_{1-m} & m \le 0, \end{cases}$$

$$[m]_+ = \begin{cases} \mathbf{e}_m & m \ge 2, \\ 2\mathbf{e}_1 & m = 1, \\ \mathbf{e}_{2-m} & m \le 0. \end{cases}$$

Only indices whose folded rows lie in $\{1, \dots, L\}$ will occur. The evaluation matrices are

$$\Phi^C_{a,p} = \phi_p(x_a), \qquad \Phi^B_{a,p} = \psi_p(x_a),$$

$$\Phi^+_{a,p} = \begin{cases} 1, & p = 1, \\ C_{p-1}(x_a), & p \ge 2. \end{cases}$$

Thus $\Phi^C[m]_C = \phi_m(x)$, $\Phi^B[m]_B = \psi_m(x)$, and $\Phi^+[m]_+ = C_{m-1}(x)$, with vector evaluation understood. In particular, $[1]_+ = 2\mathbf{e}_1$ is forced by $C_0 = 2$.

The three initial minors are the Weyl denominators

$$\begin{aligned} \Delta_C(z) &= \det[\phi_r(z_a)] = (-1)^n \prod_a z_a^{-n}(1 - z_a^2) N_1(z), \\ \Delta_B(z) &= \det[\psi_r(z_a)] = (-1)^n \prod_a z_a^{-(n-1)}(1 - z_a) N_1(z), \\ \Delta_+(z) &= \det[\chi_r(z_a)] = \prod_a z_a^{-(n-1)} N_1(z), \end{aligned} \tag{2.6}$$

where $\chi_1 = 1$ and $\chi_r = C_{r-1}$ for $r \geq 2$. These evaluations are immediate Vandermonde determinants in $z + z^{-1}$: $\phi_r/(z - z^{-1})$, $\psi_r/(z-1)$, and χ_r are monic polynomials of degree $r-1$. Inversion of a row multiplies the three denominators by -1, $-z^{-1}$, and 1, respectively.

2.2 The finite column matrices

For $\diamond \in \{C, B, +\}$, define

$$\begin{aligned} u_1^\diamond &= [k+1]_\diamond, \qquad \widetilde{u}_1^\diamond = [k]_\diamond, \\ u_r^\diamond &= [k+r]_\diamond + t^{r-1}[k+2-r]_\diamond \quad (r \geq 2), \\ \widetilde{u}_r^\diamond &= [k+r-1]_\diamond + t^{r-1}[k+1-r]_\diamond \quad (r \geq 2). \end{aligned}$$

The notation $\widetilde{u}_r^\diamond$ is used only for $\diamond = C, B$. Let $V^\star = (v_1^\star, \ldots, v_n^\star)$ be given by Table 1; every folded row there is at most $k+r$ in column r.

$\star$	Φ	$v_r^\star$	Δ
C_0	Φ^C	u_r^C	Δ_C
C_t	Φ^C	$[k+r]_C - t^r[k-r]_C$	Δ_C
BC	Φ^C	$u_r^C + \tau\widetilde{u}_r^C$	Δ_C
B_0	Φ^B	u_r^B	Δ_B
B_-	Φ^B	$[k+r]_B + \tau^{2r-1}[k+1-r]_B$	Δ_B
B_+	Φ^+	u_r^+	Δ_+
B_t	Φ^B	$u_r^B - t\widetilde{u}_r^B$	Δ_B

Table 1: Finite column data. In the BC and B_- rows, $t = \tau^2$.

Proposition 2.2. *For $n, k \geq 1$, the data of Table 1 satisfy*

$$P_{n,k}^\star = \frac{\det(\Phi V^\star)}{\Delta}.$$

Moreover, $\det V_H^\star = 1$ *for* $H = \{k+1, \ldots, k+n\}$.

Proof. Multiply (2.2) by the relevant denominator in (2.6). In the C_0 case, (2.3) gives row functions $z^{-n-k} f_r(z)$. Summing over inversions folds each row by $g(z) \mapsto g(z) - g(z^{-1})$, and these functions give $-\Phi^C u_r^C$. The n column signs cancel the factor $(-1)^n$ in Δ_C. The BC calculation multiplies the same row functions by $1 + \tau z$, giving $-\Phi^C(u_r^C + \tau\widetilde{u}_r^C)$. For C_t, replace f_r by $z^{n-r} - t^r z^{n+r}$ using (2.4); the folded column is $-\Phi^C([k+r]_C - t^r[k-r]_C)$.

For the shifted folding use $z^{-(n+k-1)} f_r(z)$ and $g(z) \mapsto g(z) - zg(z^{-1})$. This gives $-\Phi^B u_r^B$ in case B_0; multiplying by $1 - tz$ gives $-\Phi^B(u_r^B - t\widetilde{u}_r^B)$ in case B_t.

Equation (2.5) gives the B_- column in exactly the same way. Again the column signs cancel $(-1)^n$.

In case B_+, Δ_+ is invariant under each inversion. Use (2.3) and the row functions $z^{-(n+k-1)}f_r(z)$, now with $g(z) \mapsto g(z) + g(z^{-1})$. The first folded column is C_k, and the others are $C_{k+r-1} + t^{r-1}C_{k+1-r}$, exactly $\Phi^+ u_r^+$. This also checks the normalization of the first column without introducing an additional even-orthogonal alternant.

Finally, in every case the leading bracket places a 1 in row $k+r$ of column r. For $k \geq 1$ all other folded rows are strictly smaller, including those arising from nonpositive indices. Hence $V_H^\star$ is unit upper triangular. □

3 Complementary kernels and transfer

Throughout Sections 3–5, assume $n, k \geq 1$ and write

$$\begin{gathered} L = n + k, \qquad Q_0 = \{1, \dots, n\}, \qquad J = \{n+1, \dots, n+k\}, \\ H = \{k+1, \dots, k+n\}, \qquad I = \{1, \dots, k\}, \\ \omega = (-1)^{k(k+1)/2}, \qquad \zeta = (-1)^{nk}\omega. \end{gathered}$$

Subsets are always ordered increasingly. A subscript on an $n \times L$ matrix selects columns; on a matrix with L rows it selects rows.

3.1 The normalized transfer

Theorem 3.1. *Let F be a field, and let $X \in F^{n\times L}$, $Y \in F^{L\times n}$, and $U, Z \in F^{L\times k}$ satisfy $XU = 0$ and $Y^{\mathsf{T}}Z = 0$. If $\det X_{Q_0}$, $\det U_J$, $\det Y_H$, and $\det Z_I$ are nonzero, then*

$$\det(XY) = \frac{(-1)^{nk} \det X_{Q_0} \det Y_H}{\det U_J \det Z_I} \det(U^{\mathsf{T}} Z).$$

Proof. The nonzero minors give maximal rank, hence $\operatorname{col}(U) = \ker X$. The forms

$$\begin{aligned} (v_1, \dots, v_n) &\longmapsto \det[Xv_1 \ \cdots \ Xv_n], \\ (v_1, \dots, v_n) &\longmapsto \det[U \ v_1 \ \cdots \ v_n] \end{aligned}$$

induce nonzero top forms on $F^L/\operatorname{col}(U)$ and are therefore proportional. For $Q = \{q_1 < \cdots < q_n\}$ set $\epsilon(Q) = (-1)^{\sum_r (q_r - r)}$. Expansion along the last n columns gives

$$\det[U \ \mathbf{e}_{q_1} \ \cdots \ \mathbf{e}_{q_n}] = (-1)^{nk} \epsilon(Q) \det U_{Q^c}.$$

Consequently, for constants α, β independent of Q,

$$\det X_Q = \alpha \epsilon(Q) \det U_{Q^c}, \qquad \det Y_Q = \beta \epsilon(Q) \det Z_{Q^c},$$

where the second relation follows from $Y^{\mathsf{T}}Z = 0$. Cauchy–Binet gives

$$\det(XY) = \alpha\beta \sum_{|Q|=n} \det U_{Q^c} \det Z_{Q^c} = \alpha\beta \det(U^{\mathsf{T}} Z).$$

Evaluating the first relation at Q_0 and the second at H, for which $\epsilon(Q_0) = 1$ and $\epsilon(H) = (-1)^{nk}$, gives the result. □

In the finite representations, $\det \Phi_{Q_0} = \Delta$ and $\det V_H = 1$. Hence matrices D, C satisfying $\Phi D = 0$, $V^{\mathsf{T}} C = 0$, and having nonzero minors D_J, C_I give the single normalization formula

$$P^{\star}_{n,k} = \frac{(-1)^{nk}}{\det D_J \det C_I} \det(D^{\mathsf{T}} C). \tag{3.1}$$

Only the two indicated minors are needed in each application.

3.2 Three coefficient kernels

Palindromicity gives

$$\Lambda_i(-m) = -\Lambda_i(m), \qquad h_i(1-m) = h_i(m), \qquad h_i^+(2-m) = h_i^+(m). \tag{3.2}$$

In particular,

$$h_i(m) = \dot{e}_{n-i+m} + \dot{e}_{n-i-m+1}, \qquad h_i^+(m) = \dot{e}_{n-i+m} + \dot{e}_{n-i-m+2}.$$

Define $L \times k$ matrices by

$$D^C_{p,i} = (-1)^p \Lambda_i(p), \qquad D^B_{p,i} = (-1)^p h_i(p), \qquad D^+_{p,i} = (-1)^p h_i^+(p). \tag{3.3}$$

Lemma 3.2. *The three pairs satisfy* $\Phi^C D^C = 0$, $\Phi^B D^B = 0$, *and* $\Phi^+ D^+ = 0$. *Their complementary minors are* $\det D^C_J = \det D^B_J = \zeta$ *and* $\det D^+_J = 2\zeta$.

Proof. The generating identities are

$$\sum_m \Lambda_i(m) z^m = z^{-n} \dot{E}(z)(z^i - z^{-i}),$$
$$\sum_m h_i(m) z^m = z^{-n} \dot{E}(z)(z^i + z^{1-i}),$$
$$\sum_m h_i^+(m) z^m = z^{-n} \dot{E}(z)(z^i + z^{2-i}).$$

They vanish at $z = -x_a$. Pair the integer indices by $m \leftrightarrow -m$, $m \leftrightarrow 1-m$, and $m \leftrightarrow 2-m$, respectively. The first two paired sums are $\sum_{p\ge1}(-1)^p \Lambda_i(p)\phi_p(x_a)$ and $\sum_{p\ge1}(-1)^p h_i(p)\psi_p(x_a)$. In the third, the fixed index $m = 1$ gives, after division by x_a,

$$-h_i^+(1) + \sum_{p\ge2} (-1)^p h_i^+(p) C_{p-1}(x_a) = 0.$$

All three kernels vanish for $p > L$, giving the asserted kernel relations. In particular, the first row of D^+ has no extra factor 2.

For $1 \le r, i \le k$, the last k rows satisfy

$$D^C_{n+r,i} = D^B_{n+r,i} = (-1)^{n+r} \dot{e}_{i-r},$$
$$D^+_{n+r,i} = (-1)^{n+r} (\dot{e}_{i-r} + \dot{e}_{2-i-r}).$$

These blocks are upper triangular. The first two have diagonal $(-1)^{n+r}$; the third has the same diagonal except that its first entry is $2(-1)^{n+1}$, giving the stated minors. □

3.3 Folding and finite unfolding

We will fold scalar sequences $b_j(m)$, $m \in \mathbb{Z}$, into affine matrices. The three conventions are

$$\begin{aligned}
\mathcal{C}^C(b)_{p,j} &= b_j(p) - b_j(-p),\\
\mathcal{C}^B(b)_{p,j} &= (-1)^p\Big(b_j(p) + b_j(1-p)\Big),\\
\mathcal{C}^+(b)_{p,j} &= \begin{cases} -b_j(1), & p = 1,\\ (-1)^p\Big(b_j(p) + b_j(2-p)\Big), & 2 \le p \le L. \end{cases}
\end{aligned} \tag{3.4}$$

The sequences need not have finite support; the products below do.

Lemma 3.3. *For the matrices in* (3.3) *and* (3.4)*,*

$$((D^C)^{\mathsf{T}}\mathcal{C}^C(b))_{ij} = \sum_{m\in\mathbb{Z}} (-1)^m \Lambda_i(m) b_j(m), \tag{3.5}$$

$$((D^B)^{\mathsf{T}}\mathcal{C}^B(b))_{ij} = \sum_{m\in\mathbb{Z}} h_i(m) b_j(m), \tag{3.6}$$

$$((D^+)^{\mathsf{T}}\mathcal{C}^+(b))_{ij} = \sum_{m\in\mathbb{Z}} h_i^+(m) b_j(m). \tag{3.7}$$

Proof. For the first identity, pair $m = p$ with $m = -p$, using (3.2) and $\Lambda_i(0) = 0$. For the second, pair $p \ge 1$ with $1 - p \le 0$. For the third, the row $p = 1$ contributes $h_i^+(1)b_j(1)$, while $p \ge 2$ pairs with $2 - p \le 0$. Thus the fixed center is counted once, not twice. The vanishing for $p > L$ established above justifies each unfolding. □

A common bound controls all subsequent row sums. If $K > 0$ and $\mathcal{E}_r = 0$ outside $[0, N]$, a row with entries

$$\mathcal{E}_{n-i+j-Ky} + \sigma\mathcal{E}_{n+i+j-Ky-\delta}, \qquad 1 \le j \le k, \quad \delta \in \{0, 1, 2\},$$

vanishes unless

$$\left\lceil \frac{n - i + 1 - N}{K} \right\rceil \le y \le \left\lfloor \frac{n + i + k - \delta}{K} \right\rfloor.$$

Indeed, outside this interval every index is above N or below 0; here $2i - \delta \ge 0$. Take $N = 2n$, or $2n + 1$ for $\widehat{h}$. Thus, if $A_{ij} = \sum_y w_j(y) F_i(j - Ky)$, finite row multilinearity gives $\det A = \mathfrak{J}_K[w, F; \mathbb{Z}^k]$. The even-sublattice case will use a parity projection instead.

Every weight has exponent $\frac{K}{2}y^2 - (j - \eta)y$, with $\eta \in \{0, \frac{1}{2}, 1\}$, $K/2 \ge k - \eta$, and $j - \eta \ge 0$. It is nonnegative for every integer y. In case B_0 it is an integer because $y^2 - y$ is even; in the square-root cases twice this exponent is the exponent of τ. The resulting affine expressions are therefore polynomial in t or τ, even though reflection relations will be proved in the corresponding rational function fields.

4 The four ordinary affine kernels

Four specializations have no doubled complementary block. The two unshifted foldings share one reflection, and the two shifted foldings share another.

4.1 The pair C_0 and BC

Proposition 4.1. *Equations* (1.6) *and* (1.8) *hold for* $n, k \geq 1$.

Proof. Take $K = 2k + 2$ for C_0 and $K = 2k + 1$ for BC, with $t = \tau^2$ in the latter case. For $1 \leq j \leq k$ define

$$b_j(m) = \begin{cases} t^{Kq^2/2+jq}, & m = j + Kq, \quad q \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{K}, \end{cases}$$
$$c_j(m) = b_j(m) - b_j(-m),$$

and let $C = \mathcal{C}^C(b)$. Shifting q gives $b_j(m + K) = t^{m+K/2} b_j(m)$, hence

$$c_j(-m) = -c_j(m), \qquad c_j(m) + t^{m-K/2} c_j(K - m) = 0. \tag{4.1}$$

The second relation follows by applying the quasiperiodicity to $b_j(K - m)$ and $b_j(m - K)$; it holds also outside the residue classes.

For C_0, the relation at $m = k + 1$ gives $c_j(k + 1) = 0$, and at $m = k + r$ it gives $c_j(k + r) + t^{r-1} c_j(k + 2 - r) = 0$. Since the odd extension agrees with the bracket $[m]_C$, these are precisely $(V^{C_0})^{\mathsf{T}} C = 0$. For BC, the first column gives $c_j(k + 1) + \tau c_j(k) = 0$. For $r \geq 2$, put $h = k + r$. The four terms in its finite column split as

$$(v_r^{BC})^{\mathsf{T}} C_{\bullet,j} = \Big(c_j(h) + \tau^{2h-K} c_j(K - h)\Big) + \tau\Big(c_j(h - 1) + \tau^{2(h-1)-K} c_j(K - h + 1)\Big) = 0.$$

Thus the second kernel relation holds in both cases.

Because $K > 2k$, the residues j and $-j$ are disjoint and the first k rows give $C_I = I_k$. By (3.5) and $y = -q$,

$$(D^C)^{\mathsf{T}} C = A \operatorname{diag}((-1)^1, \ldots, (-1)^k),$$
$$A_{ij} = \sum_{y \in \mathbb{Z}} (-1)^{Ky} t^{Ky^2/2 - jy} \Lambda_i(j - Ky).$$

The sign $(-1)^{Ky}$ is 1 for C_0 and $(-1)^y$ for BC. Since $\det D_J^C = \zeta = (-1)^{nk}\omega$, (3.1) gives $P_{n,k}^{\star} = \det A$. Finite row multilinearity proves the two identities. □

4.2 The pair B_0 and B_-

Proposition 4.2. *Equations* (1.3) *and* (1.4) *hold for* $n, k \geq 1$.

Proof. Take $K = 2k + 1$ for B_0 and $K = 2k$ for B_-, with $t = \tau^2$ in the latter case. Define

$$b_j(m) = \begin{cases} (-1)^{Ky} t^{Ky^2/2 - (j-1/2)y}, & m = j - Ky, \quad y \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{K}, \end{cases}$$

and put

$$\gamma_j(m) = (-1)^m\Big(b_j(m) + b_j(1 - m)\Big), \qquad C = \mathcal{C}^B(b).$$

Changing y to $y-1$ gives $b_j(m+K) = (-1)^K t^{m+(K-1)/2} b_j(m)$. Applying this relation to the two reflected terms yields

$$\gamma_j(1-m) = -\gamma_j(m), \qquad \gamma_j(m) = -t^{m-(K+1)/2}\gamma_j(K+1-m). \tag{4.2}$$

For example, both $b_j(K+1-m)$ and $b_j(m-K)$ acquire the factor $(-1)^K t^{(K+1)/2-m}$ from $b_j(1-m)$ and $b_j(m)$. The parity in the definition of γ_j leaves the minus sign in (4.2) for either parity of K.

For B_0, the fixed index $m = k+1$ in the second reflection has $\gamma_j(k+1) = 0$; the remaining columns are killed by $\gamma_j(k+r) + t^{r-1}\gamma_j(k+2-r) = 0$. For B_-, it gives instead $\gamma_j(k+r) + \tau^{2r-1}\gamma_j(k+1-r) = 0$. The first reflection makes evaluation against $[m]_B$ agree with $\gamma_j(m)$, so $V^{\mathsf{T}}C = 0$ in both cases.

For $1 \le p, j \le k$, $b_j(p) = \delta_{p,j}$, whereas $b_j(1-p) = 0$ because $1 \le j+p-1 \le 2k-1 < K$. Thus $C_I = \operatorname{diag}((-1)^1, \dots, (-1)^k)$ and $\det C_I = \omega$. The common pairing (3.6) gives

$$((D^B)^{\mathsf{T}}C)_{ij} = A_{ij} := \sum_{y\in\mathbb{Z}} (-1)^{Ky} t^{Ky^2/2-(j-1/2)y} h_i(j-Ky).$$

Now (3.1) and $\det D_J^B = \zeta$ give $P^\star_{n,k} = \det A$. For odd K the affine sign is $(-1)^y$; for even K it is absent. Finite multilinearity proves the stated identities. □

5 Fixed classes and adjacent residues

The remaining cases have one doubled complementary minor. We treat separately its three sources: a fixed residue, a fixed center, and the overlap of adjacent residues.

5.1 The fixed residue in C_t

Proposition 5.1. *Equation* (1.7) *holds for* $n, k \ge 1$.

Proof. Set $K = 2k$ and define

$$b_j(m) = \begin{cases} (-1)^q t^{kq^2+jq}, & m = j+2kq, \quad q \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{2k}. \end{cases}$$

Let $c_j(m) = b_j(m) - b_j(-m)$ and $C = \mathcal{C}^C(b)$. The quasiperiodicity is now $b_j(m+2k) = -t^{m+k} b_j(m)$, so the calculation leading to (4.1) gives

$$c_j(-m) = -c_j(m), \qquad c_j(m) = t^{m-k} c_j(2k-m).$$

At $m = k+r$ this annihilates the column $[k+r]_C - t^r[k-r]_C$; hence $(V^{C_t})^{\mathsf{T}}C = 0$.

Unlike the ordinary cases, the residues k and $-k$ coincide. Their contributions add rather than cancel:

$$c_k(k+2kq) = 2(-1)^q t^{kq(q+1)} \quad (q \in \mathbb{Z}).$$

For $j < k$ the two residues remain disjoint. Consequently, $C_I = \mathrm{diag}(1, \dots, 1, 2)$ and $\det C_I = 2$. This includes $k = 1$, when the block consists of the single entry 2.

Write

$$a_{ij}(y) = t^{ky^2 - jy} \Lambda_i(j - 2ky),$$
$$A^+_{ij} = \sum_y a_{ij}(y), \qquad A^-_{ij} = \sum_y (-1)^y a_{ij}(y).$$

The pairing (3.5), with $y = -q$, gives $(D^C)^{\mathsf{T}} C = A^- \,\mathrm{diag}((-1)^1, \dots, (-1)^k)$. Thus the fixed-residue minor and (3.1) yield $P^{C_t}_{n,k} = \frac{1}{2} \det A^-$.

It remains to identify the even-sublattice expression, not the full lattice sum. The parity projector and finite multilinearity give

$$\sum_{y \in \mathbb{Z}^k_{\mathrm{even}}} \det[a_{ij}(y_i)] = \tfrac{1}{2}\Big(\det A^+ + \det A^-\Big).$$

In the last column, $y \mapsto 1 - y$ preserves $ky^2 - ky$ and sends $k - 2ky$ to its negative. Since Λ_i is odd, $a_{i,k}(1 - y) = -a_{i,k}(y)$. This fixed-point-free involution makes the last column of A^+ zero. The parity projection therefore equals $\frac{1}{2} \det A^-$, proving the identity. □

5.2 The fixed center in B_+

The sequence used here will also supply the adjacent-residue reflection in the next subsection. For $1 \le j \le k$, set

$$b_j(m) = \begin{cases} (-1)^y t^{ky^2 - (j-1)y}, & m = j - 2ky, \quad y \in \mathbb{Z}, \\ 0, & m \not\equiv j \pmod{2k}, \end{cases} \tag{5.1}$$
$$\Gamma_j(m) = (-1)^m \Big(b_j(m) + b_j(2 - m)\Big).$$

Lemma 5.2. *For every integer m,*

$$b_j(m + 2k) = -t^{m+k-1} b_j(m),$$
$$\Gamma_j(2 - m) = \Gamma_j(m),$$
$$\Gamma_j(m) = -t^{m-k-1} \Gamma_j(2k + 2 - m). \tag{5.2}$$

In particular, $\Gamma_j(k + 1) = 0$.

Proof. The first relation follows from $y \mapsto y - 1$ in (5.1); the second is the defining symmetry. For the last, the first relation gives

$$b_j(2k + 2 - m) = -t^{k+1-m} b_j(2 - m), \qquad b_j(m - 2k) = -t^{k+1-m} b_j(m).$$

Since m and $2k + 2 - m$ have the same parity, adding these relations proves (5.2). Its fixed index $m = k + 1$ has eigenvalue -1 and hence vanishes. □

Proposition 5.3. *Equation* (1.5) *holds for* $n, k \geq 1$.

Proof. Take $C = \mathcal{C}^+(b)$ for (5.1). The normalized center row is $C_{1,j} = -b_j(1)$, so

$$[1]_+^{\mathsf{T}} C_{\bullet,j} = 2C_{1,j} = -2b_j(1) = \Gamma_j(1).$$

Together with $\Gamma_j(2-m) = \Gamma_j(m)$, this shows that every bracket $[m]_+$ evaluates to $\Gamma_j(m)$, including the fixed center. Lemma 5.2 therefore gives $(V^{B_+})^{\mathsf{T}} C = 0$: the first column evaluates to $\Gamma_j(k+1)$ and the others to $\Gamma_j(k+r)+t^{r-1}\Gamma_j(k+2-r) = 0$.

The first k rows are $C_I = \operatorname{diag}((-1)^1, \ldots, (-1)^k)$. Indeed, $b_j(1) = \delta_{j,1}$; for $2 \leq p \leq k$, $b_j(p) = \delta_{p,j}$ and the reflected congruence would require $2ky = j+p-2 \in \{1, \ldots, 2k-2\}$, which is impossible. Thus $\det C_I = \omega$. The factor 2 occurs instead in $\det D_J^+ = 2\zeta$, whose first diagonal coefficient is $\dot{e}_0 + \dot{e}_0$ by Lemma 3.2.

By (3.7),

$$((D^+)^{\mathsf{T}} C)_{ij} = A_{ij}^{B_+} := \sum_y (-1)^y t^{ky^2-(j-1)y} h_i^+(j-2ky).$$

In this pairing the center contributes exactly $h_i^+(1)b_j(1)$; its coefficient is not doubled. Formula (3.1) now gives $P_{n,k}^{B_+} = \frac{1}{2}\det A^{B_+}$, and finite row multilinearity proves the result. □

5.3 Adjacent residues in B_t

This case uses the same coefficient matrix D^B as B_0 and B_-. Its different row kernel satisfies $\widehat{h}_i(m) = h_i(m) + h_i(m-1)$, directly from $\ddot{e}_r = \dot{e}_r + \dot{e}_{r-1}$. Keep b_j and Γ_j from (5.1) and put

$$\begin{aligned} a_j(m) &= b_j(m) + b_j(m+1), \\ c_j(m) &= (-1)^m \Big(a_j(m) + a_j(1-m) \Big), \\ \gamma_j(m) &= c_j(m) - t c_j(m-1). \end{aligned}$$

Thus a_j is supported on the two adjacent residues j and $j-1$, and $c_j(1-m) = -c_j(m)$.

Lemma 5.4. *The adjacent-residue sequences satisfy*

$$c_j(m) = \Gamma_j(m) - \Gamma_j(m+1), \tag{5.3}$$

$$\gamma_j(m) = (1+t)\Gamma_j(m) - \Gamma_j(m+1) - t\Gamma_j(m-1), \tag{5.4}$$

$$\gamma_j(m) = -t^{m-k-1}\gamma_j(2k+2-m). \tag{5.5}$$

In particular, $\gamma_j(k+1) = 0$.

Proof. Expanding the definitions proves (5.3), and subtracting $tc_j(m-1)$ gives (5.4). Let $T = 2k+2$ and $d = m-k-1$. Apply (5.2) at $m, m+1, m-1$ to the latter identity:

$$\begin{aligned} \gamma_j(m) &= -t^d \Big((1+t)\Gamma_j(T-m) - t\Gamma_j(T-m-1) - \Gamma_j(T-m+1) \Big) \\ &= -t^d \gamma_j(T-m). \end{aligned}$$

The fixed-index assertion follows at $m = k+1$. □

Proposition 5.5. *Equation* (1.9) *holds for* $n, k \geq 1$.

Proof. Set $C = \mathcal{C}^B(a)$, so $C_{p,j} = c_j(p)$. Evaluation against the first finite column gives

$$([k+1]_B - t[k]_B)^{\mathsf{T}} C_{\bullet,j} = \gamma_j(k+1) = 0.$$

For $r \geq 2$, the four terms in $u_r^B - t\tilde{u}_r^B$ give $\gamma_j(k+r) + t^{r-1}\gamma_j(k+2-r) = 0$ by (5.5). Hence $(V^{B_t})^{\mathsf{T}} C = 0$.

The first complementary block contains the overlap that fixes the normalization. For $1 \leq p, j \leq k$, only $y = 0$ contributes, and

$$a_j(p) = \delta_{p,j} + \delta_{p,j-1}, \qquad a_j(1-p) = \delta_{p,1}\delta_{j,1}.$$

The second equality follows because the two congruences require $2ky = j + p - 1$ or $2ky = j + p - 2$; only the latter can occur, at $p = j = 1$. Therefore

$$C_{p,j} = (-1)^p\Big(\delta_{p,j} + \delta_{p,j-1} + \delta_{p,1}\delta_{j,1}\Big) \quad (1 \leq p, j \leq k).$$

This block is upper bidiagonal, with diagonal $-2, 1, -1, 1, \ldots$, and hence $\det C_I = 2\omega$. There is no doubling in $\det D_J^B = \zeta$.

Finally, the common pairing (3.6) and the definition of $\widehat{h}_i$ give

$$\begin{aligned}((D^B)^{\mathsf{T}} C)_{ij} &= \sum_m h_i(m)\Big(b_j(m) + b_j(m+1)\Big) \\ &= \sum_y (-1)^y t^{ky^2-(j-1)y}\Big(h_i(j-2ky) + h_i(j-1-2ky)\Big) \\ &= \sum_y (-1)^y t^{ky^2-(j-1)y}\widehat{h}_i(j-2ky).\end{aligned}$$

The normalization formula yields one half of this determinant, and finite multilinearity proves Theorem B. □

6 Width zero and classical specializations

6.1 The normalized empty rectangle

Proposition 6.1. *For every* $n \geq 1$, *all seven normalized width-zero values are* 1. *Consequently, the five non-prefactored identities in Theorem A hold also at* $k = 0$.

Proof. The symmetric-group reduction (2.2) is valid at $k = 0$. The alternant and row-folding calculation in Proposition 2.2 therefore gives

$$\nu_\star(0) P^\star_{n,0} = \frac{\det(\Phi V^\star|_{k=0})}{\Delta},$$

without using the complementary-kernel transfer. Relative to the initial basis defining Δ, these column transformations are upper triangular. Their diagonal coefficients in column r are

Case	Diagonal coefficient	Product
C_0, BC, B_0	1	1
C_t, B_t	$1+t^r$	$(-t;t)_n$
B_-	$1-\tau^{2r-1}$	$(\tau;\tau^2)_n$
B_+	$1+t^{r-1}$	$(-1;t)_n$

For example, the B_t columns are $(1+t)\psi_1$ and

$$(1+t^r)\psi_r - (t+t^{r-1})\psi_{r-1} \qquad (r \geq 2),$$

while the BC columns are ϕ_1 and

$$\phi_r + (\tau - \tau^{2r-1})\phi_{r-1} - \tau^{2r-2}\phi_{r-2} \qquad (r \geq 2).$$

For B_+ the first column is $C_0 = 2$ relative to $\chi_1 = 1$, so its diagonal is indeed $1 + t^0 = 2$. Each product in the table equals $\nu_\star(0)$, proving the claim. The five affine expressions without an external half-factor are 1 by the empty-determinant convention. □

These are rational-function identities before cancellation; any removable singularities of the displayed normalization factors are understood after cancellation. In particular, the conclusion does not require a nonvanishing numerical value of every Pochhammer factor.

6.2 The specialization $t = 0$

The classical rectangular dual Jacobi–Trudi formulas are recovered in three finite foldings. They are the symplectic and orthogonal determinants in [2, Theorem 2.1]; see also the lattice-path proofs in [3].

Corollary 6.2. *For $n, k \geq 1$, with $1 \leq a, r \leq n$ on the left and $1 \leq i, j \leq k$ on the right,*

$$\frac{\det[\phi_{k+r}(x_a)]}{\Delta_C(x)} = \det[\dot{e}_{n-i+j} - \dot{e}_{n+i+j}], \tag{6.1}$$

$$\begin{aligned}\frac{\det[\psi_{k+r}(x_a)]}{\Delta_B(x)} &= \det[\dot{e}_{n-i+j} + \dot{e}_{n+i+j-1}] \\ &= \tfrac{1}{2}\det[\ddot{e}_{n-i+j} + \ddot{e}_{n+i+j-1}],\end{aligned} \tag{6.2}$$

$$\frac{\det[C_{k+r-1}(x_a)]}{\Delta_+(x)} = \tfrac{1}{2}\det[\dot{e}_{n-i+j} + \dot{e}_{n+i+j-2}]. \tag{6.3}$$

The seven finite representations specialize to the corresponding left sides: C_0, C_t, BC to (6.1); B_0, B_-, B_t to (6.2); and B_+ to (6.3).

Proof. At $t = 0$ (and $\tau = 0$ in the square-root cases), every finite column in Table 1 reduces to $[k + r]_\diamond$. This identifies the finite ratios. In the affine formulas for C_0, B_0, B_+, B_t, every nonzero shift has a strictly positive parameter exponent, leaving only $y = 0$ in each row and giving the displayed dual determinants. For C_t one should instead use its finite ratio: at $j = k, y = 1$ the weight exponent is zero, so nonzero shifts do not disappear entrywise. The already proved parity identity identifies its specialized even-sublattice sum with the same symplectic determinant. □

AI use declaration. Artificial intelligence was used to assist in exploring proof strategies and developing preliminary code for numerical experiments. All results were independently verified by the authors. The authors assume responsibility for all content.

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